

UNIT 5

ADVANCED PROPULSION SYSTEMS

Chemical rockets use the energy stored in the propellants to create a hot gas, which then becomes the working fluid in a heat engine, and is expelled, generating thrust. There is an elegant simplicity in this triple function of the propellant and its combustion products, which is reflected in the simple nature of the rocket engine. There is, however, a fundamental limitation which results from combining the functions of working fluid and energy source: no more energy can be put into the rocket than is contained in the propellant flowing into the engine. This means that the power output of the rocket is rigidly defined by the chemical energy and flow rate of the propellant. The exhaust velocity and thrust are defined by the thermodynamic relationships in Chapter 2, and there is no possibility of exceeding these values. As has previously been pointed out, the arrival of the space age was dependent on stretching the ability of chemical rockets to the limit, through multi-staging, and on engines that perform very close to their theoretical best. More ambitious space programmes—manned missions to Mars, for example could be achieved with the same technology, but would require a very large effective mass ratio because of the velocity increment involved. Moreover, all the necessary propellant would need to be raised to Earth orbit. It would be preferable if somehow more propulsive power could be extracted from the propellant, and the exhaust velocity could be increased beyond the 4.5 km/s that is the best available from chemical rockets.

Principles of Electric Propulsion

As mentioned above, the basic principle of electric propulsion is to apply electrical energy to the propellant from an external power source. This can be done in several ways. The simplest is to heat the propellant with a hot wire coil, through which an electric current passes. This elementary approach, used in some commercial thrusters, is very successful. More energy can be delivered from the electric current if an arc is struck through the propellant, which generates higher temperatures than the resistive approach and produces a higher exhaust velocity. Finally, electric or magnetic fields can be used directly to accelerate propellant ions to very high velocities, producing the highest exhaust velocity of all. These ion thrusters, and Hall Effect thrusters are seen as the most promising for deep space applications, and they are already coming into commercial use for station keeping and interplanetary propulsion. While for a chemical rocket the link between energy supply and propellant simplifies analysis, for electrical propulsion the power supply introduces free parameters for which we have to make estimates when deriving expected vehicle performance. Electric power can come from a battery, solar panels or an onboard nuclear or solar generator, each of which has its own advantages and disadvantages.

What is important, from the vehicle performance point of view, is the power-to-mass ratio-W/kg. In most cases the power does not diminish with progress through the flight, while the mass of propellant decreases in the familiar way as the vehicle accelerates. This is in direct contrast to the chemical rocket, in which both the propellant and the available energy decrease together. Using these ideas, simple estimates of vehicle performance can be produced.

ELECTRIC THRUSTERS

Electric thrusters can be divided into two broad categories: those that use electricity to heat the propellant, which emerges as a neutral gas, and those which use electric or magnetic fields to accelerate ions. The functional form and analysis of these two classes differ.

Electrothermal Thrusters- These are the type of rockets in which the thrust produced is purely based on the principle of thermodynamics. Since the thrust produced by the electrical rockets are very less, they are usually termed as thrusters rather than rockets.

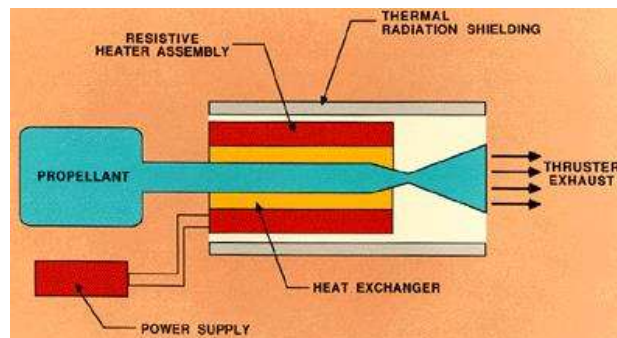
Resistive method - The simplest way is to heat the propellant with a hot wire through which electric current passes. Though this method seems very elementary it has been used in some commercial thrusters and are very successful.

Electric Arc Method - In case of more energy requirements, using the electric current an electric arc can be struck and when the propellants are fed through this electric arc, the propellants are heated up and passed through a nozzles to accelerate it further to create thrust. This method generates higher temperatures and higher exhausts velocities than the resistive method.

3) Electro-Static Rockets – These rockets use electric and magnetic fields to produce thrust.

In this method electric or magnetic fields are used to directly accelerate the propellants to very high velocities producing the highest exhaust velocities of all. The ion thrusters and hall effect thrusters comes under this category and are seen as the most promising for deep space missions, interplanetary missions and station keeping missions.

Electro thermal Rockets-Resistojet Rockets



Schematic of a Resistojet Thruster

Electro-thermal rockets use both electrical energy and thermal energy or specifically, the principles of thermodynamics to heat up the propellant and expand the exhaust gas in a suitable nozzle to create thrust. In this case, the rocket consists of a wire or a filament that heats up due to resistive heating and the heat that is generated is used for generating thrust and hence the name Resistojet. The same principle happens in household geysers or industrial geysers which are used to heat the water. The basic Resistojet Rocket consist of propellant storage tank (ammonium, hydrogen, nitrogen, or hydrazine decomposition product gases have been used as propellants), heating chamber consisting of heating coils made up of tungsten or platinum, heat exchanger and heat shields, Power supply and a nozzle with high expansion ratio connected to the heating chamber.

During working, the heater coil or the heater assembly will first be heated though the supplies of electric current from the on-board power supply system. The current supplied and hence the maximum gas temperature achieved in the heating chamber should always be less than the maximum serviceable temperature of the filament. Then as the high pressure gas from the propellant tank is made to pass through the heating chamber, the gas molecules will get heated up and gets converted into high temperature high pressure exhaust gases that will be expelled through the nozzle. So using lower molecular weight propellants will produce significant exhaust velocities.

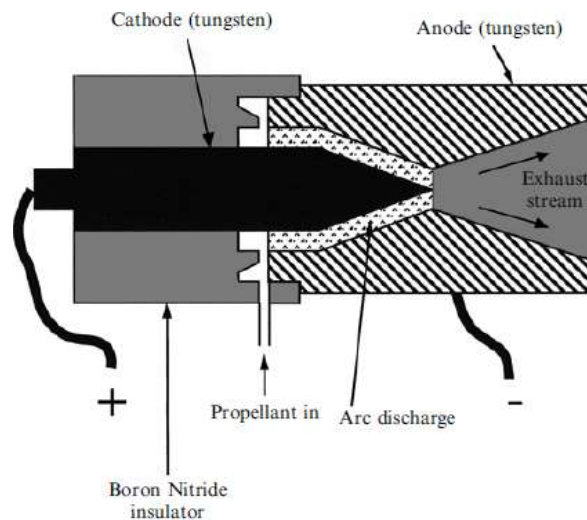
To achieve high velocities, the pressure and temperature of the gas entering the nozzle need to be high. As the propellant is stored at high pressure, the losses in pressure are not as pronounced as the losses in temperature. Since gas is a bad conductor of heat, only a thin layer in close contact with the heater will become hot. To overcome this issue, a multichannel heat exchanger is used to enhance the heat radiation to as much as gas volume possible so that this we should make sure that there is efficient heating of the gas. Next, the heated wire invariably radiates some amount of heat to the chamber walls and in doing so rate of heat transfer to the gas will be poor that results in low temperature and pressure when the gas enters the nozzle. And when this occurs the total thrust and exhaust velocities are reduced. To prevent this, low mass radiation shields made up of concentric foils are used or in some cases active cooling techniques are used that can also help in regenerative heating as done on the chemical rocket. The specific impulse I_{sp} for resistojet thrusters is 150 to 700 I_{sp} range, efficiency levels between 35% and 90%, and thrust between 5 and 5,00 mN (0.5N). The thrust produced by these electric propellant engines will be equivalent to the force created by a falling paper or feather. Since these engine operate in vacuum, even milli-newtons of thrust produced for a long time can accelerate the vehicles to a large exhaust velocities close to 5000m/s translating close to Mach 14.

Advantages

- Fuel load reduction
- Cost savings in launch vehicle option for lighter spacecraft
- Reduction of integration and testing costs by eliminating the use of hazardous propellants
- Low volume and power budgets

Electric Arc Thrusters - Arcjet Thrusters

This is another technique which uses electrical and thermal energies to propel space vehicles especially in vacuum. Arcjet thrusters consist of a centrally located cathode surrounded by an anode. The pointed cathode rod is supported in a boron nitride insulator which also holds the anode. The anode is shaped to create the gap across which the arc is struck and through which the propellant flows. The anode is further shaped as a nozzle which expands the gases to high exhaust velocities.



Schematic of Arcjet Thruster

Using the on-board electric power supply system, a steady DC potential of 200-300V high voltage electric field is struck between the cathode and anode while the propellant is injected between the two electrodes. Liquid Hydrogen (Ammonia and Hydrazine are also used), the propellant used in arcjet thrusters when passed through this high strength electric arc will be heated and result in the generation of high temperature ions. These rapidly moving ions will again recombine as they move downstream of the nozzle and when it enters the nozzle; it will be expanded to cause high velocities. In arcjet thrusters, the temperature of the propellants can be raised between 10,000 K and 20,000 K. The maximum temperature attained in arcjet thrusters is always higher compared to the resistojet thrusters where the maximum temperature is limited by the operating temperature of the wire filament. Higher temperatures in an arcjet thruster directly translate to significantly higher exhaust velocities close to 20000m/s. The indirect application of these extremely high temperatures through an electric current provides insulation for operational elements, and the expulsion pattern of the superheated gas contributes to the insulating effect. Comparatively large Isp ranges, between 280 and 2,300, give significant advantages over resistojet propulsion methods, while a comparable thrust range of 50 to 5,000 mN. However, the energy conversion efficiency of the arcjet thruster will be between 30% and 50% which is lesser than the resistojet thrusters. Additional problem with arcjet thrusters is that they are prone to electrode erosion and massive power requirements.

Electro-Static Propellant Rockets

In the previous section, we understood that by using Resistojet Thrusters, maximum exhaust velocities close to 5000m/s can be achieved. Then by switching to arcjet thrusters, maximum exhaust velocities can be further increased to 20km/s. But for interplanetary and interstellar space missions, the required exhaust velocities are much beyond that could be achieved using resistojet and arcjet thrusters. If we wish to exceed the exhaust velocities using electrical heating of propellants, then it is necessary to abandon thermodynamic effects and act directly upon the atoms the propellant by using pure electrostatic or electromagnetic fields. This means that the propellant has to be ionised. This is already happening in case of arcjet thrusters, but this again recombines within the nozzle to create neutral atoms that are later expanded. But the processes of ionization, recombination is a technically complicated and nuisance process. To simplify the process, the propellant needs to be ionised first and then using electric and magnetic fields the ions can be directly accelerated producing very high bulk velocities needed. Using this technique theoretical exhaust velocities close to 50000m/s can be achieved but however, practical thrusters have produced exhaust velocities beyond 30000m/s.

Ion Propulsion

The concept of ion propulsion is very simple as compared to the other versions of electric propulsion. Ion propulsion or ion thrusters use electro-static and electro-magnetic principles to accelerate ions to very high exhaust velocities and produce thrust. To achieve this, in an ion thruster, the ions are produced first and then accelerated to high velocities. This can be explained with the schematic below.

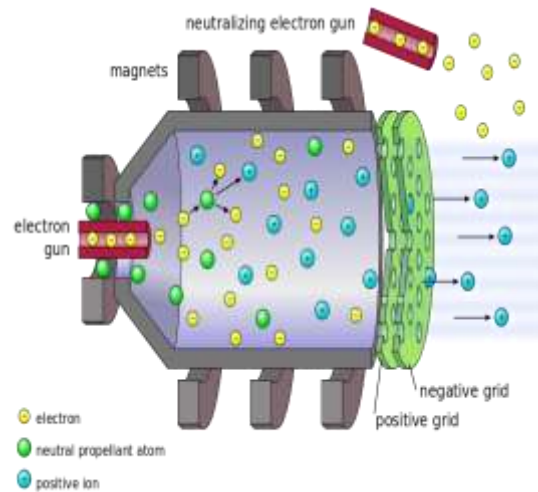


Fig: Illustration of a typical ion propulsion system

In ion thrusters, the process of ionization is a very important in which a neutral atom will be converted into a positively and negatively charged ions and electrons. Ionization is achieved by bombarding the propellant gas with high energy fast moving electron beam. When high energy electrons are bombarded to a neutral gas, the neutral atoms will be split into positive ions and electrons. The gas consisting of both positive ions and electrons is called as plasma and plasma is usually affected by the electric and magnetic fields.

In an ion thruster, the propellant is made to enter the ionization chamber and the propellant when it enters will in the form of neutral state. Xenon is the commonly used propellant as it has high atomic mass resulting in large ion density and gets easily ionized. The ionization chamber is facilitated with radial electric fields. Along with the propellant, the electrons from a thermionic emitter will be made to simultaneously enter the chamber. The electrons that enter the chamber will be accelerated by the radial field and reach energies of several tens of electron volt which is enough to ionise the neutral propellant atom by collisions. To increase the path length of the electrons and to ensure that they encounter as many as neutral atoms possible, an axial magnetic field is provided which makes the neutral atoms move in a spiral path. This is done to increase the number of ions produced through collisions that increases the efficiency of ionization process. This process is similar to atomization process in a traditional combustion chamber of a rocket or an aircraft. At the end of the ionization chamber two accelerator grids at a distance of 1-2mm and with thousands of precisely drilled holes are provided. Through diffusion, the positive ions are forced to move towards the first grid and they pass through it under the influence of small voltage. Later as it approaches the negative grid, a huge potential difference across the negative grid will accelerate the negative ions to very high velocities. There is no need for a nozzle to create thrust because the motions of the ions are ordered and not chaotic. Here the thermodynamic relations to calculate the thrust and performance parameter are not valid.

Plasma Thrusters

The low thrust and very high exhaust velocity of ion engines are a disadvantage for many applications where the efficiency of electric propulsion would be beneficial. These disadvantages are attributable to the fact that only the positive ions contribute to thrust, and the ion current is limited to a low value by the space charge effect. If the ion flow could be increased beyond the space charge limit, then a much more versatile engine could be developed. It would have higher thrust and a somewhat lower exhaust velocity more in keeping with the requirements of a wide range of missions. For many years it has been the dream of engine designers to produce a practical plasma thruster. The principle is simple. An ionised gas passes through a channel across which are maintained orthogonal electric and magnetic fields (see Figure 6.14). The current, carried by electrons and ions, which develops along the electric field vector, interacts with the magnetic field to generate

a propulsive force along the channel. The force acts in the same direction for both electrons and ions, and so the whole plasma is accelerated; and to a first order, the accelerating force is not limited by the density of the plasma, so there is no limit analogous to the space charge limit. The gas does not, in fact, need to be completely ionised; even a few percent of ions is sufficient, because they transfer their energy to the neutral gas molecules by collision. Energy lost by the ions in this way is immediately restored by the electric and magnetic fields.

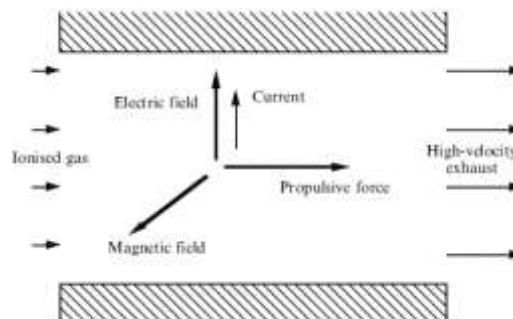


Fig: Operation of a Plasma Thruster

Such a device would have exactly the characteristics required: a higher mass flow giving higher thrust, and exhaust velocities in the 10-20 km/s range, ideal for many missions. The principle is analogous to that of a linear electric motor, where the current flowing through the plasma is represented by the current in the armature, and the accelerating force acts in the same way. The process is, however, nowhere near as simple as this: the transverse current generates its own magnetic field, the gas is heated, and the ions are acted upon by electric and magnetic fields which they themselves generate. A combination of thermodynamics and electromagnetic theory is required to predict the outcome.

This science magnetoplasmadynamics— was gleefully seized upon by theoreticians, with no fewer than 16 interrelations between parameters. At the same time many experimental devices were made and tested, but nothing emerged which could be used in space-until one of those occurrences arose which restore faith in experimental science. Workers experimenting with arc-jets at very low gas pressures (millibars) observed that currents of the order of 3,000 amps could be made to flow with no erosion of the electrodes, and with exhaust velocities of up to 100 km/s and an efficiency of 50%. This was obviously a magnetoplasma dynamic effect, but the mechanism was obscure. The experimental results were undeniable, and this remained the best hope for the development of a practical plasma thruster until the end of the Cold War.

Hall Effect Thrusters

Over the years, Russia developed a plasma thruster based on the Hall effect, and implemented it on more than 100 satellites. This device has now become available worldwide, and is simple and practical. The Hall thruster belongs to the family of magnetoplasma dynamic devices, and has been shown to share the practical properties of the low-pressure arc-jet described above-and it works. However, it was only in the Russian space programme that it had been brought to a practical and space-qualified form. A Hall effect thruster is a small rocket engine that uses a powerful magnetic field to accelerate a low density plasma and so produce thrust. The Hall effect thruster, also called a plasma thruster, is a form of electrostatic propulsion, which in turn is a form ion propulsion (a category of electric space propulsion). Like gridded ion engines, such as XIPS, Hall thrusters are classified as electrostatic thrusters. Both utilize an inert gas, commonly xenon, as a propellant.

Electrons are generated by a hollow cathode (negative electrode) at the downstream end of the thruster. The anode (positive electrode) or "channel" is charged to a high potential by the thruster's power supply. The electrons are attracted to the channel walls and accelerate in the upstream direction. As the electrons move toward the channel, they encounter a magnetic field produced by the thruster's powerful electromagnets. This high-strength magnetic field traps the electrons, causing them to form into a circling ring at the downstream end of the thruster channel. The Hall thruster gets its name from this flow of electrons, called the Hall current.

The propellant, which consists of an inert gas such as xenon or krypton at low pressure, is injected into the thruster's channel. Since Hall thrusters use inert gas for propellant, there is no risk of explosion as there is with chemical rockets. Some of the trapped electrons in the channel collide with the propellant atoms, creating ions.

When the propellant ions are generated, they experience the electric field produced between the channel (positive) and the ring of electrons (negative) and accelerate out of the thruster, creating an ion beam. The thrust is generated from the force that the ions impart to the electron cloud. This force is transferred to the magnetic field, which, in turn, is transmitted to the magnetic circuit of the thruster. The electrons are highly mobile and attracted to the ions in the beam, causing an equal amount of electrons and ions to leave the thruster at the same time. This enables the thruster to remain overall electrically neutral.

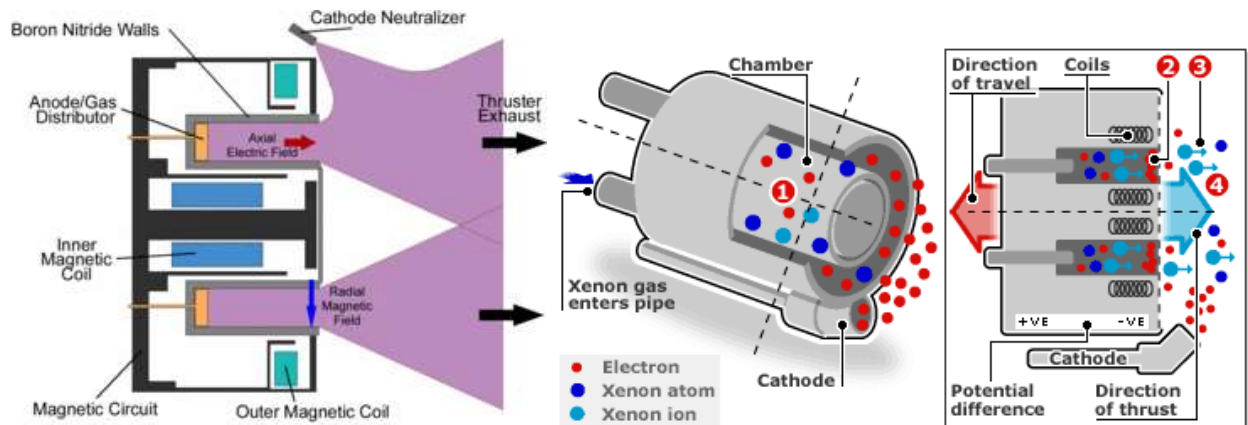


Fig: Working of a Hall effect thruster and Movement of the xenon ions and atoms and electrons in the thruster

Radio Frequency Thruster

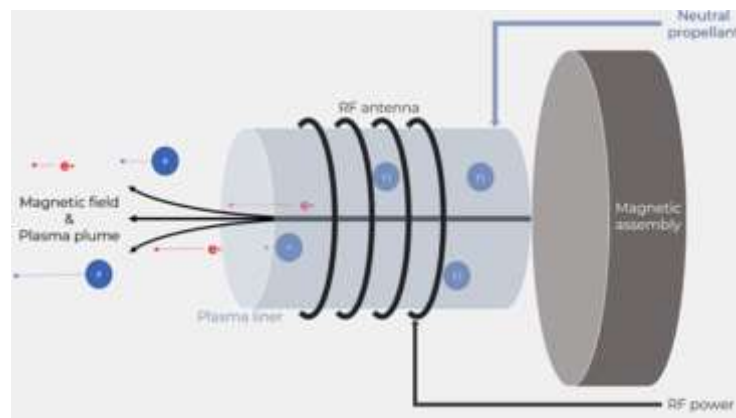
Most high power-and hence high thrust-systems use electrodes, of one kind or another, to generate the current in the gas that provides the ions and hence the thrust. These always erode in the discharge, being worse for high power systems. Several attempts have been made to increase the power input by using microwaves to provide the internal energy source. The simplest device is analogous to the electrothermal thruster: microwaves are used to heat the gas in a 'combustion' chamber connected to a de-Laval nozzle, which converts the hot gas into an exhaust stream. The 'combustion' chamber is in fact a microwave cavity designed to set up standing electromagnetic waves that heat the gas by accelerating electrons, which in turn ionise the propellant, allowing higher microwave induced currents to flow in the gas and heat it to propellant temperatures. Some laboratory thrusters of this type have been made.

A much more complex scheme is the variable specific impulse device, called VASIMIR; it uses radio frequency electromagnetic fields to ionise and accelerate the plasma: The process here begins with the gas, hydrogen and helium mixed, being exposed to the electromagnetic radiation from an RF antenna that ionises most of the atoms. It then passes into a cavity with strong magnetic fields, where cyclotron resonance is excited by a high-power RF generator, resonance here gives a very high efficiency of power transfer from the RF field to the gas. The electrons and ions oscillate within the plasma and raise its overall temperature to a very high value.

Temperature here is somewhat different from temperature in an un-ionised gas, because the ions and electrons behave somewhat differently from neutral molecules at high temperatures. Nevertheless, the assemblage of ions and electrons behaves somewhat like a gas at high temperature and pressure. It then enters a magnetic nozzle, which behaves like a conventional nozzle, in that it allows the hot, high-pressure plasma to expand, and so to generate a high-velocity exhaust stream, and thrust, in the conventional way. The difference is that the nozzle has no mechanical presence at all; it is made up from a carefully shaped, strong, static, magnetic field. The electrons and ions in the plasma are forced to travel along the diverging magnetic field lines, so that the plasma expands as it emerges from the engine.

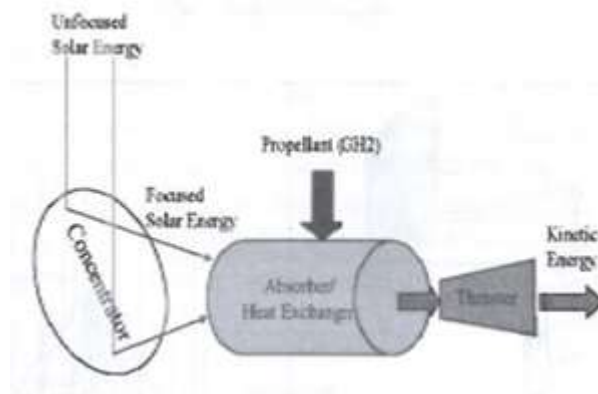
This is the basic scheme, but for the VASIMIR engine, unionised gas is injected into an outer, mechanical nozzle, which surrounds the magnetic field nozzle. This gas is heated by contact with the ionised gas in the magnetic nozzle, and adds to the thrust. Variability in the thrust is achieved by changing the size of the magnetic nozzle throat, simply by changing the field. At the same time, the amount of heating and the propellant flow rates can be separately adjusted to change the exhaust velocity. In many ways this device is analogous to a chemical rocket, but it has the advantages of electric propulsion. It is claimed that this device

can operate at very high powers, and can be optimised for high-thrust missions. So far, only laboratory demonstrations of the processes involved have been made.



Solar Thermal Propulsion

The concept of solar thermal propulsion stems from the better known solar-dynamic power system. This propulsion system has been proposed as an alternative means of rocket propulsion for orbit transfer and planetary missions. In this system, solar energy is accumulated by an absorber and is used to burn propellant during the thrust phase.



Solar thermal propulsion concept

In general, solar thermal propulsion utilises solar light to heat a propellant up to 2000 K. The gaseous propellant is then fed through a conventional rocket nozzle to produce thrust. A solar propelled rocket only needs a means by which it can capture solar energy. No electrical generator is needed. The thrust of the engine is dependent on the surface area of the solar collector (concentrators and mirrors) and the local intensity of the solar radiation. Light is collected by a parabolic reflector and then focused onto a blackbody cavity to generate a high internal temperature. Here, heat is transferred to the propellant to produce thrust. The best performance can be achieved with hydrogen because of its low molecular weight. Concentrators are used to collect solar light and to focus it into a receiver/absorber. Large collector surfaces in the range of several square meters are needed to produce the required thermal energy. A desired solar power input will be about 1350 W/m². This is typically realised by the primary concentrators and thus, they are considered to be the most technologically demanding part of the concentrator system. The function of the absorber/receiver is to absorb and transfer the energy of the concentrated solar radiation to the propellant. A heat exchanger is employed in the heating of the propellant. The transfer of energy can be achieved either continuously or after accumulation.

Solar thermal propulsion performance is in between that of conventional propulsion and ion drive propulsion. For indirect solar heating, the design cannot achieve specific impulses of more than 900 s. This is because there are some limitations to the temperature that the heat exchanger material can withstand. For direct solar heating, direct heat absorption allows higher propellant temperatures and hence has higher specific impulses of approximately 1200s. However, only indirect solar thermal heating has been experimentally tested. This was carried out by the United States Air Force Rocket Propulsion Laboratory using small-scale models.

Solar Sails

Back in the 1600s, a physicist Johannes Kepler postulated the presence of solar breeze from his observations of comet tails. That led to the formulation of a miraculous method of space travel that does not require fuel and is capable of lasting forever. However it was not until the 1970s that the construction of a solar sail at the Jet Propulsion Laboratory (JPL) actually took place for a rendezvous with comet Halley. While the mission never did take place, the designs conceptualised from this mission laid the foundation for further research.

Solar sails are born from the concept that light can generate force to propel spacecraft. Light consists of packets of photons which carry momentum. Photons impinging on the solar sails will undergo reflection, and a resultant force causes the solar sail to move. At a distance of 1 AU (astronomical unit), the photons exert a force of 9 N/km², or roughly one thousandth of the weight of two paper clips at the earth's surface. This force then grows exponentially with distance to produce a large impulse. Solar sails employ a technique known as tacking to control their orbits. The force exerted by sunlight on a solar sail comprises of the forces from the incoming and the reflected sunlight which are at right angles to each other. Together, they produce a net force vector that always points away from the sun. Should this force be angled against the direction of travel, the orbit will shrink towards the sun. To expand the orbit, the force should be angled in the direction of travel.

Radioisotope thermoelectric generators

A radioisotope thermoelectric generator, or RTG, uses the fact that radioactive materials (such as plutonium) generate heat as they decay into non-radioactive materials. The heat used is converted into electricity by an array of *thermocouples* which then power the spacecraft. A thermocouple is a device which converts thermal energy directly into electrical energy. Basically, it is made of two kinds of metal that can both conduct electricity. They are connected to each other in a closed loop. If the two metals are at different temperatures, an electric potential will exist between them. When an electric potential occurs, electrons will start to flow, making electric current. This is a much different technique than that used by nuclear power stations on Earth. That process is called *fission*, and gets very high efficiency rates by literally "splitting" unstable radioactive materials (such as uranium) into smaller parts. Fission generates very large amounts of heat, but is much more complex and not as reliable as simply using the heat produced by radioactive decay. Basically, fission gives you a huge release of energy and uses fuel rapidly. An RTG gives a steadier and much smaller amount of energy.

Radioisotope Thermoelectric Generators (RTG), also called Radioisotope Power Systems (RPS) are power production systems most often used in long-distance space travel and remote areas on Earth. They produce electricity by the heat emitted from decaying radioactive isotopes. RTGs are most famous for their use on NASA's Cassini probe, Voyager probes, New Horizons.^[2] and Curiosity rover.^[3] See below for three dimensional models.

RTGs can be broken down into three critical components; the radioactive isotope, the thermocouples, and the thermal insulation.

Isotope

The radioactive isotope is at central to the RTGs design. This nuclear source provides the heat from which the rest of the design produces electricity. Most designs use Plutonium-238, Strontium-90, or Americium-241 for their power density, half-life, and radioactive shielding requirements. Because the half-lives of the isotopes are strategically chosen, RTGs provide power for decades. For example, Plutonium-238, the most common and famous isotope used on NASA RTGs has a half-life of approximately 90 years.^[4] Each isotope has advantages and drawbacks, but usually one of these three isotopes will be deposited in pellets at the center of the RTG to which they can then decay and emit heat to the surrounding components.

Thermocouple

The thermocouple, discovered in 1821 by Thomas Johann Seebeck^[6] is a device that can produce electricity by differences in surrounding temperature. When two different metals are connected and there is a temperature difference between them, a magnetic field is produced. This magnetic field can then be used to produce a current, or flow of electrons, and a resulting voltage and power output. This is the exact fundamental basis used in thermocouples on an RTG. The waste heat produced by decaying isotopes will heat one metal, while

the ambient temperature of the RTG and deep space will keep the other metal cool. Just as before, these metals are connected and electricity is produced.

Thermal Regulation

The final main component of an RTG is the thermal insulation. Especially important for deep space probes, the device must be able to maintain a certain temperature to keep electronics and moveable parts operating properly. The RTG is covered with many layers of insulation and then a hard outer shell to protect the critical workings of the device. However, not all the waste heat is needed to keep the spacecraft running. As such, most RTGs have radiator fins. Regulating temperatures on an RTG is critical to its proper function and operation. Specific design principles are needed for this task, and temperature regulation makes up a bulk of the outer casings and layers.

NUCLEAR PROPELLANT ROCKETS

Nuclear energy is always associated with the transformation of atomic particles within the nucleus of atoms and these atomic transformations are always associated with the liberation of enormous amounts of heat energy. In Nuclear propellant rockets, this liberated heat energy is utilized to heat to a working fluid, usually liquid hydrogen, which subsequently can be expanded in a nozzle and thus accelerated to high ejection velocities (6000 to 10,000 m/sec).

Three different types of nuclear energy sources have been investigated to be employed for rocket propulsion. They are Nuclear Fission, Radioactive Isotope source decay and Nuclear Fusion processes. A number of different concepts have been studied. To date none have been tested and many concepts are not yet feasible or practical. Concerns about an accident with the inadvertent spreading of radioactive materials in the earth environment and the high cost of development programs have to date prevented a renewed experimental development of a large nuclear rocket engine. Unless there are some new findings and a change in world attitude, it is unlikely that a nuclear rocket engine will be developed or flown in the next few decades.

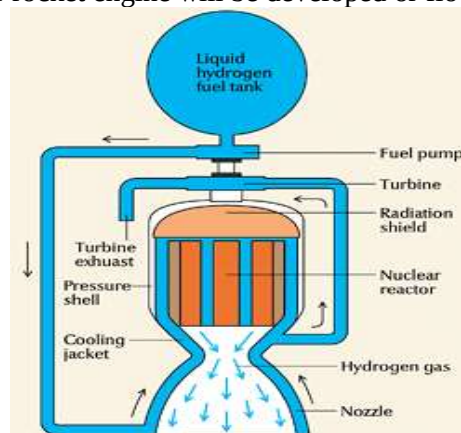


Fig : Working of a nuclear rocket engine

In the nuclear fission reactor rocket, heat can be generated by the fission of uranium in the solid reactor material and subsequently transferred to the working fluid. The nuclear fission rocket is primarily a high-thrust engine (above 40,000 N) with specific impulse values up to 900 sec. Fission rockets were designed and tested in the 1960s. Ground tests with hydrogen as a working fluid culminated in a thrust of 980,000 N at a graphite core nuclear reactor level of 4100 MW with an equivalent altitude-specific impulse of 848 sec and a hydrogen temperature of about 2500 K. There were concerns with the endurance of the materials at the high temperature (above 2600 K) and intense radiations, power level control, cooling a reactor after operation, moderating the high energy neutrons, and designing lightweight radiation shields for a manned space vehicle. In recent years there has been renewed interest in nuclear fission rocket propulsion primarily for a potential manned planetary exploration mission. Studies have shown that the high specific impulse (estimated in some studies at 1100 sec) allows shorter interplanetary trip transfer times, smaller vehicles, and more flexibility in the launch time when planets are not in their optimum relative position.